

Assignment 2: Coding Basics

Student Name

OVERVIEW

This exercise accompanies the lessons/labs in Environmental Data Analytics on coding basics.

Directions

1. Rename this file `<FirstLast>_A02_CodingBasics.Rmd` (replacing `<FirstLast>` with your first and last name).
2. Change “Student Name” on line 3 (above) with your name.
3. Work through the steps, **creating code and output** that fulfill each instruction.
4. Be sure to **answer the questions** in this assignment document.
5. When you have completed the assignment, **Knit** the text and code into a single PDF file.
6. After Knitting, submit the completed exercise (PDF file) to Canvas.

Basics, Part 1

1. Generate a sequence of numbers from one to 55, increasing by fives. Assign this sequence a name.
2. Compute the mean and median of this sequence.
3. Ask R to determine whether the mean is greater than the median.
4. Insert comments in your code to describe what you are doing.

```
#1. builds a sequence using 1 and 55 as the start and end numbers, increases by 5's
sequence1 <- seq(1,55,5)
sequence1
```

```
## [1] 1 6 11 16 21 26 31 36 41 46 51
```

```
#2. creates objects that are the mean and median values of sequence1
mean_sequence1 <- mean(sequence1)
mean_sequence1
```

```
## [1] 26
```

```
median_sequence1 <- median(sequence1)
median_sequence1
```

```
## [1] 26
```

```
#3. Returns the logical value of the statement -- mean is not greater than median of sequence1
mean_sequence1 > median_sequence1
```

```
## [1] FALSE
```

Basics, Part 2

5. Create three vectors, each with four components, consisting of (a) student names, (b) test scores, and (c) whether they are on scholarship or not (TRUE or FALSE).
6. Label each vector with a comment on what type of vector it is.
7. Combine each of the vectors into a data frame. Assign the data frame an informative name.
8. Label the columns of your data frame with informative titles.

```
#5,#6 creating three vectors
a <- c('Forest','River','Sky', 'Aldeberon') #Student names, character
b <- c(30,82,93,98) #Student test scores, numeric
c <- c(FALSE,TRUE,FALSE,TRUE) #Students on scholarship?, logical
#7 combining vectors into a data frame
student_info <- data.frame(a,b,c)
#8 labeling columns
names(student_info) <- c('Student Names','Student Scores','On Scholarship')
student_info
```

```
##   Student Names Student Scores On Scholarship
## 1      Forest          30          FALSE
## 2       River          82           TRUE
## 3        Sky          93          FALSE
## 4    Aldeberon          98           TRUE
```

9. QUESTION: How is this data frame different from a matrix?

Answer: This dataframe can mix and match data types, while matrices can't

10. Create a function with one input. In this function, use `if...else` to evaluate the value of the input: if it is greater than 50, print the word "Pass"; otherwise print the word "Fail".

```
passfail_function <- function(x){
  if(x>50){
    print('pass')}
  else{
    print('Fail')}
}
```

11. Create a second function that does the exact same thing as the previous one but uses `ifelse()` instead of `if...else`.

```
passfail_function_ifelse <- function(x){
  ifelse(x>50, print('Pass'), print('Fail'))
}
```

12. Run both functions using the value 52.5 as the input

```
passfail_function(52.5)
```

```
## [1] "pass"
```

```
passfail_function_ifelse(52.5)
```

```
## [1] "Pass"
```

```
## [1] "Pass"
```

```
#both functions work!
```

13. Run both functions using the **vector** of student test scores you created as the input. (Only one will work properly...)

```
##Sorry, for the below I put in R chunks above...
```

```
#10. Create a function using if...else
```

```
#11. Create a function using ifelse()
```

```
#12a. Run the first function with the value 52.5
```

```
#12b. Run the second function with the value 52.5
```

```
#13a. Run the first function with the vector of test scores
```

```
#passfail_function(student_info)
```

```
#13b. Run the second function with the vector of test scores
```

```
passfail_function_ifelse(student_info)
```

```
## [1] "Pass"
```

```
## [1] "Fail"
```

```
##      Student Names Student Scores On Scholarship
```

```
## [1,] "Pass"          "Fail"          "Fail"
```

```
## [2,] "Pass"          "Pass"          "Fail"
```

```
## [3,] "Pass"          "Pass"          "Fail"
```

```
## [4,] "Pass"          "Pass"          "Fail"
```

14. QUESTION: Which option of `if...else` vs. `ifelse` worked? Why? (Hint: search the web for “R vectorization”)

Answer: The ‘ifelse’ worked, because the ifelse function is an vectorized function that handles the entire dataset in bulk!

NOTE Before knitting, you’ll need to comment out the call to the function in Q13 that does not work. (A document can’t knit if the code it contains causes an error!)